Test Protocol

P/N: PICUS Q

S/N: 0X-LH-222-017 + 0X-PS-220-022

DATE: 13.2.2024

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1. Introduction

The measured values shown in this document were generated with the laser system intended for the NBI during final inspection.

2. Output Power

The average power of the Signal and Idler fiber outputs depending on the wavelength settings is shown in **Figure 1**. The output power varies across the tuning range. For this custom version of the Picus Q, the Signal and the Idler is transported by a PM980 patch cord with FC/APC termination.

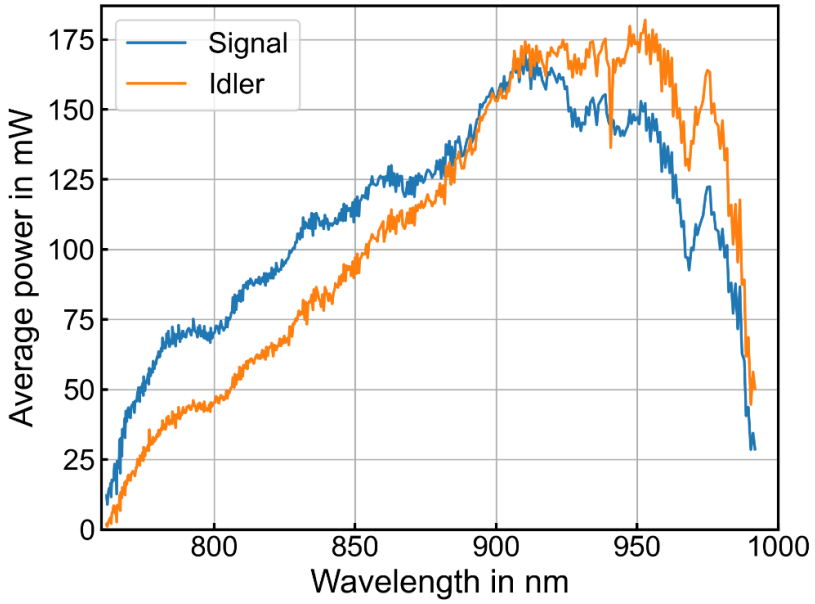


Figure 1: Average output power from the Signal and Idler fiber output depending on the central wavelength setting.

The output power of the Idler fiber is shown in **Figure 2,** depending on its central wavelength.

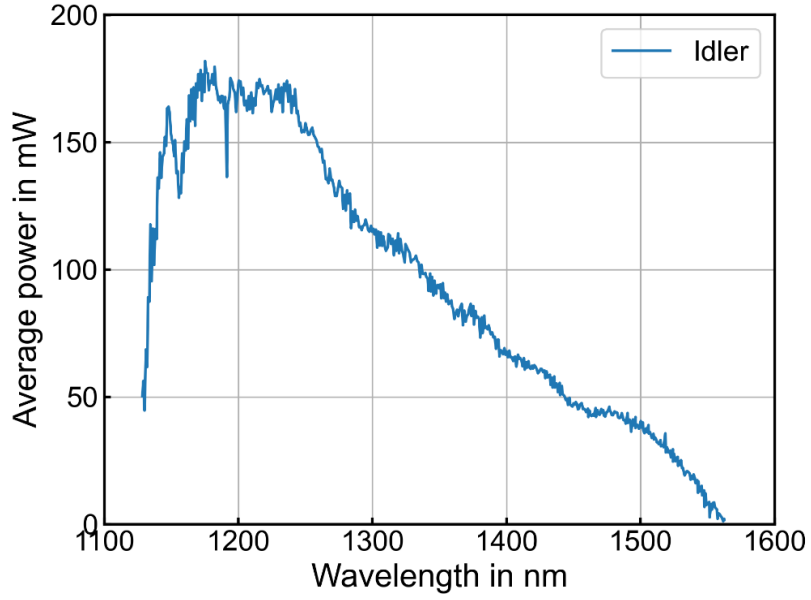


Figure 2: Average output power from the Idler fiber output depending on the central wavelength of the Idler.

The tuning range of the filtered Signal output is narrower, than the output of the Signal. Engaging the Wavelength Lock ensures a match between the filter setting and the actual output of the Signal, as its central wavelength is depending on the ambient conditions. **Figure 3** shows the output power of the filtered Signal fiber, depending on the wavelength setting, with the Wavelength Lock engaged.

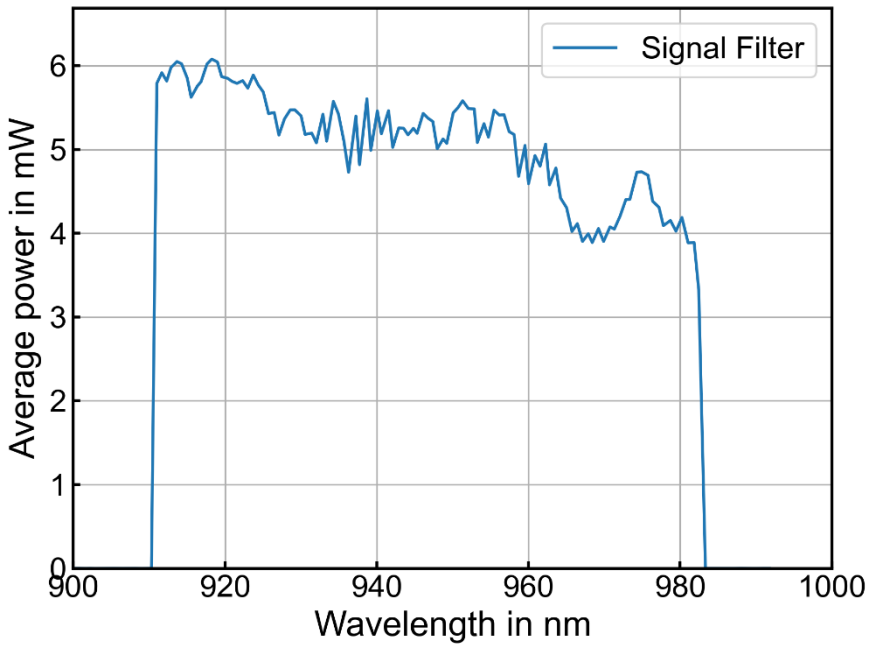


Figure 3: Average output power from the filtered Signal fiber output depending on the central wavelength setting.

The accessible tuning range is summarized in **Table 1**. Setting values below or above the stated limits results in a “bounds error”.

Table 1: Summary of the tuning limits.

	Wavelength Signal (nm)	**Wavelength Idler (nm)**	**Wavfilter (bits)**
Upper wavelength limit	991	1129	1300
Upper Filter limit	981	1141	1440
Lower Filter limit	911	1236	2455
Lower wavelength limit	762	1561	6200

3. Pulse Durations

The pulse duration has been measured with a home-built interferometric autocorrelator, calibrated intrinsically by the distance of the interference fringes. The measured intensity autocorrelation shown in **Figure 4** was calculated from interferometric traces and is representative for the performance over the tuning range (plotted in **Figure 5**).

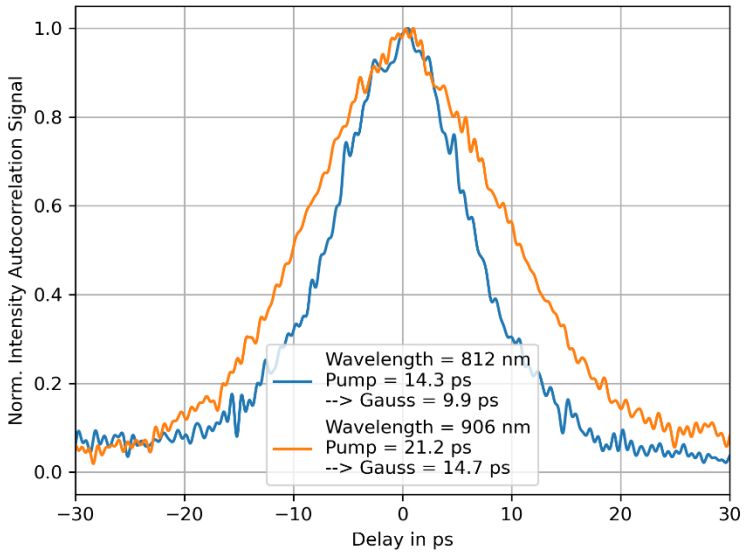


Figure 4: Representative intensity autocorrelation trace of the fiber output.

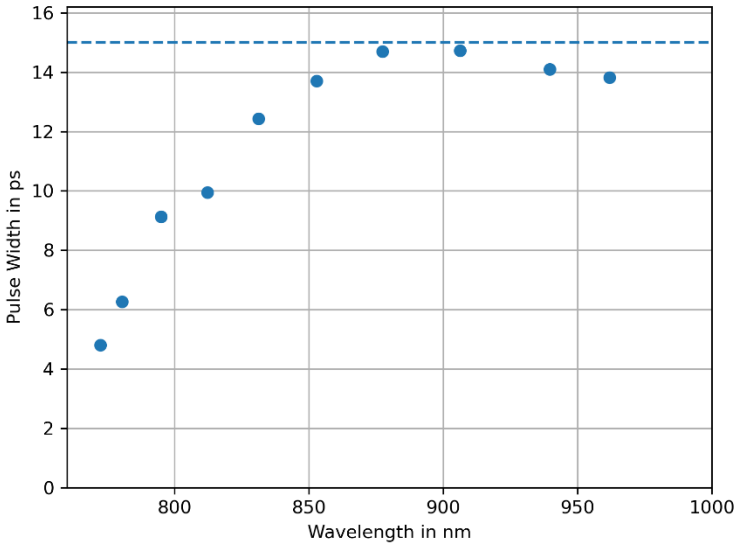


Figure 5: Estimated pulse durations, retrieved from autocorrelation traces assuming a Gaussian pulse shape, at wavelength settings across the tuning range.

4. Output Spectra

The output spectra in **Figure 6** and **Figure 7** were taken after the output fiber. The spectra are exemplary representatives for the complete tuning range. Note the cascaded signals for longer output wavelengths than 900 nm. These cascades are orders of magnitude weaker than the main peak.

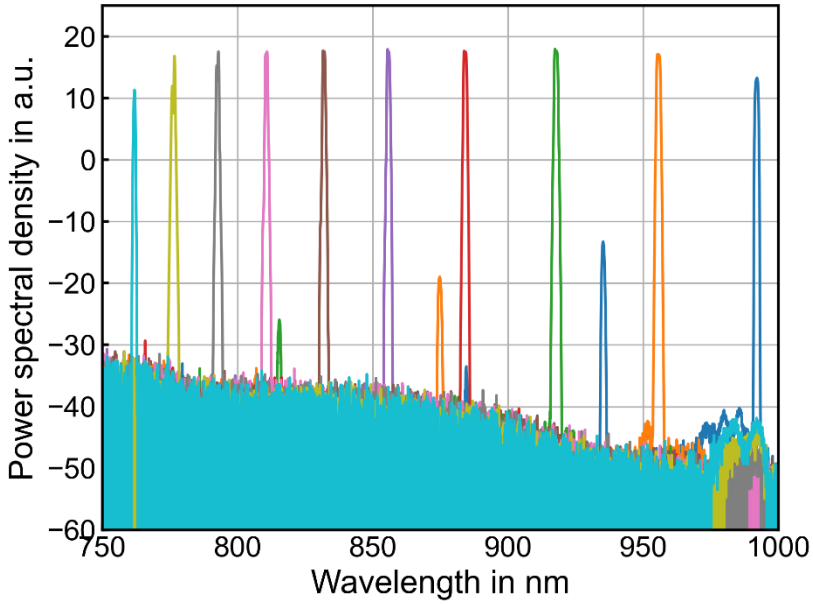


Figure 6: Signal output spectra at various wavelength settings.

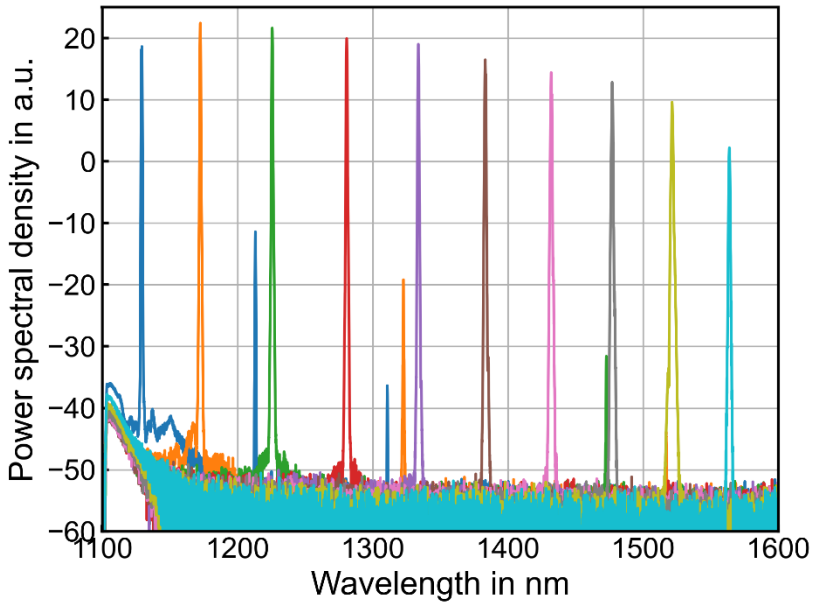


Figure 7: Idler output spectra at various wavelength settings.

Figure 8 shows representative spectra of the filtered Signal output fiber.

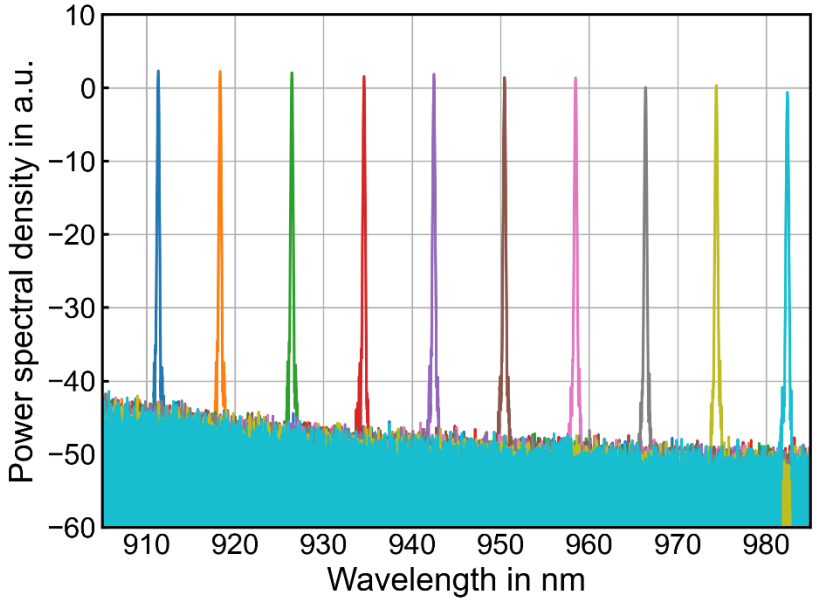


Figure 8: Filtered Signal output spectra at various wavelength settings.

The spectral bandwidth of the output pulses is shown in **Figure 9** as a function of the center wavelength. The spectral bandwidth of the Signal increases for longer center wavelengths, while the bandwidth of the Idler decreases.

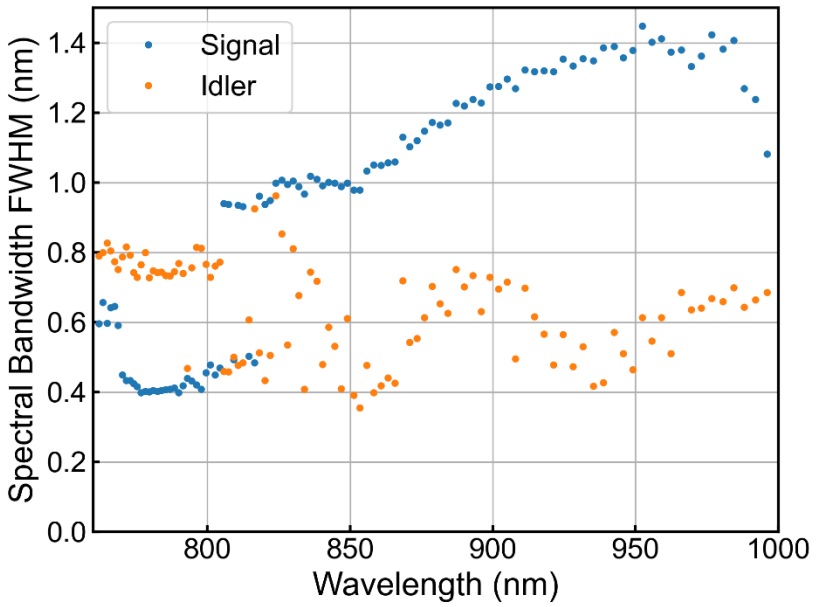


Figure 9: Spectral bandwidth as a function of the center wavelength.

5. Repetition frequency

While tuning the center wavelength of the output pulses, the repetition frequency of the pulse train changes accordingly. The dependency of the repetition frequency on the wavelength is depicted in **Figure 10**.

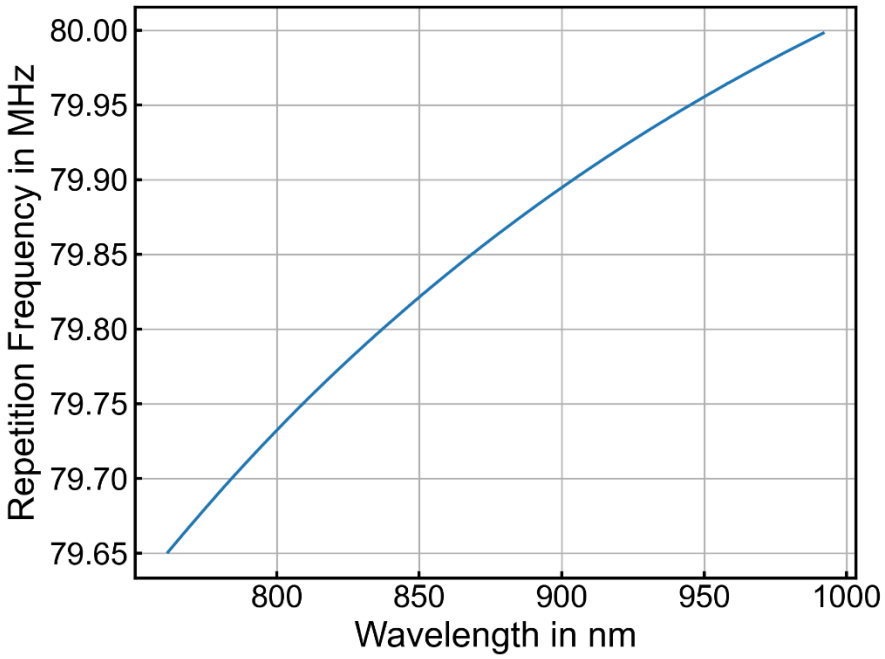


Figure 10: Repetition frequency depending on the wavelength setting.

6. Warm-up Time

The system becomes operable and emits light within seconds after starting the system. A change of 5-10% of the output specifications over 2 h is possible until stabilized. It is recommended to use the Picus Wavelength Lock with the internal calibration during warm-up to account for wavelength drifts.